

BOMING ZHANG

Ph.D. Candidate in Computer Science | AI/ML Education | Intelligent Tutoring Systems
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PROFILE

Computer science Ph.D. candidate and AI education practitioner with experience building intelligent tutoring systems, classroom LLM integrations, computer vision/video analysis workflows, and project-based machine learning curricula. Strong background translating advanced AI concepts into usable learning tools for students, instructors, and public-sector partners.

EDUCATION

Ph.D. in Computer Science University of Massachusetts Amherst, Amherst, MA	Expected Dec 2026
Ed.S. in Mathematics, Science, & Learning Technologies (MSLT) UMass Amherst, Amherst, MA	Expected Dec 2026
M.S. in Computer Science University of Massachusetts Amherst, Amherst, MA <i>Award: Bay State Master's Program</i>	May 2021
B.A. in Computer Science Hampshire College, Amherst, MA <i>Award: Innovator Award for Problem Solving</i>	May 2019

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, R, Swift, JavaScript/React, Clojure, Django
AI / Data Tools: PyTorch, YOLOv5, ffmpeg, PySceneDetect, decord, data visualization, educational data mining
Systems & Databases: AWS, Docker, MongoDB, MySQL, Git, web dashboards, video-processing pipelines
Applied Areas: Machine learning, computer vision, LLM classroom integration, intelligent tutoring systems

EXPERIENCE

School Instructor & Curriculum Designer X School - Teach high school students advanced machine learning concepts through project-based coding labs, lectures, and applied demonstrations. - Design classroom workflows that integrate large language models into instruction while emphasizing responsible use, prompt literacy, and reflection. - Implement LLM-powered learning systems, including student-facing dialogue interfaces and teacher dashboards for monitoring progress and engagement.	May 2026 - Present
Software Engineer / Research Assistant Center for Knowledge and Communication, UMass Amherst, Amherst, MA - Develop and deploy the MathSpring intelligent tutoring system on AWS to improve scalability, reliability, and student engagement. - Create interactive multimedia math problems for K-12 students aligned with curriculum standards. - Design educational games and learning activities for the Wearable Learning Cloud Platform.	Apr 2020 - Present
UMass Intern - Digital Evidence Lab Office of the Attorney General, Massachusetts - Integrated scene detection and GPU-accelerated video processing into a digital evidence workflow using PySceneDetect and decord. - Built automated pipelines to extract scene changes from large video files and export timestamped frames to CSV for analysis. - Created a video lookup GUI, added support for new video formats, and delivered live tutorials and remote demos for AG Office staff.	Jan 2024 - Aug 2024
On-Campus Instructor iDTech, Cambridge, MA - Delivered AI-focused curriculum through hands-on coding lessons, lectures, assessments, and individualized student support. - Maintained a 100% positive feedback rate from students and parents.	Jun 2023 - Aug 2023
Data Science Fellow - Scholarship America Project Center for Data Science, UMass Amherst, Amherst, MA - Cleaned and analyzed 300K+ student records to evaluate scholarship allocation patterns. - Identified racial and gender disparities in scholarship outcomes and contributed to public-facing data reports.	May 2022 - Aug 2022
Undergraduate Research Assistant MIT Poggio Lab, Center for Brains, Minds + Machines, Cambridge, MA - Developed R modules for neural decoding analysis and reproducible research workflows. - Deployed computational tasks to AWS and created Docker images for portable research environments.	May 2018 - Aug 2018

SELECTED PROJECTS & PUBLICATIONS

The Routledge Companion to Digital Media and Children | Publication Contributor
- Contributed to a chapter summarizing research on children learning coding and its developmental benefits.

Five Colleges DataFest | Best Business Insight Award, 2018
- Recognized for translating data analysis into actionable business insight during a collaborative data competition.

No-Chalk Teaching Project | Educational Technology Initiative
- Introduced an electronic whiteboard drawing solution to replace chalk-based teaching at No. 166 Middle School in China.